

DS 4440

Beyond Linear Models

Last time: Linear Regression from an "ML" perspective - i.e., $\nabla \mathcal{L}$ / SGD

$$\nabla_w \mathcal{L} = -2x^T y + 2 \underbrace{x^T x}_{d \times d} w_{dx_1}$$

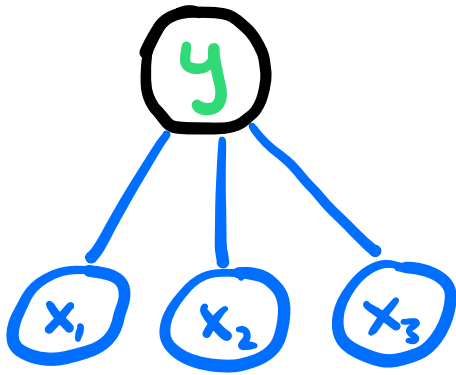
Analytic Solution

$$w^* = (x^T x)^{-1} x^T y$$

Gradient Descent

$$\hat{w}^{(t+1)} \leftarrow \hat{w}^{(t)} - \underbrace{\alpha}_{\text{learning rate}} \nabla_w \mathcal{L}(\underbrace{x, y}_{\text{usually batched}} | w^{(t)})$$

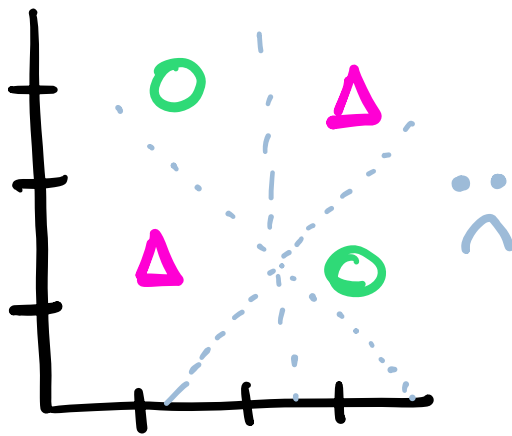
So far we have assumed



$$y = f(w_0 + w_1 x_1 + w_2 x_2 + w_3 x_3)$$

This implies a line or hyperplane that separates classes.

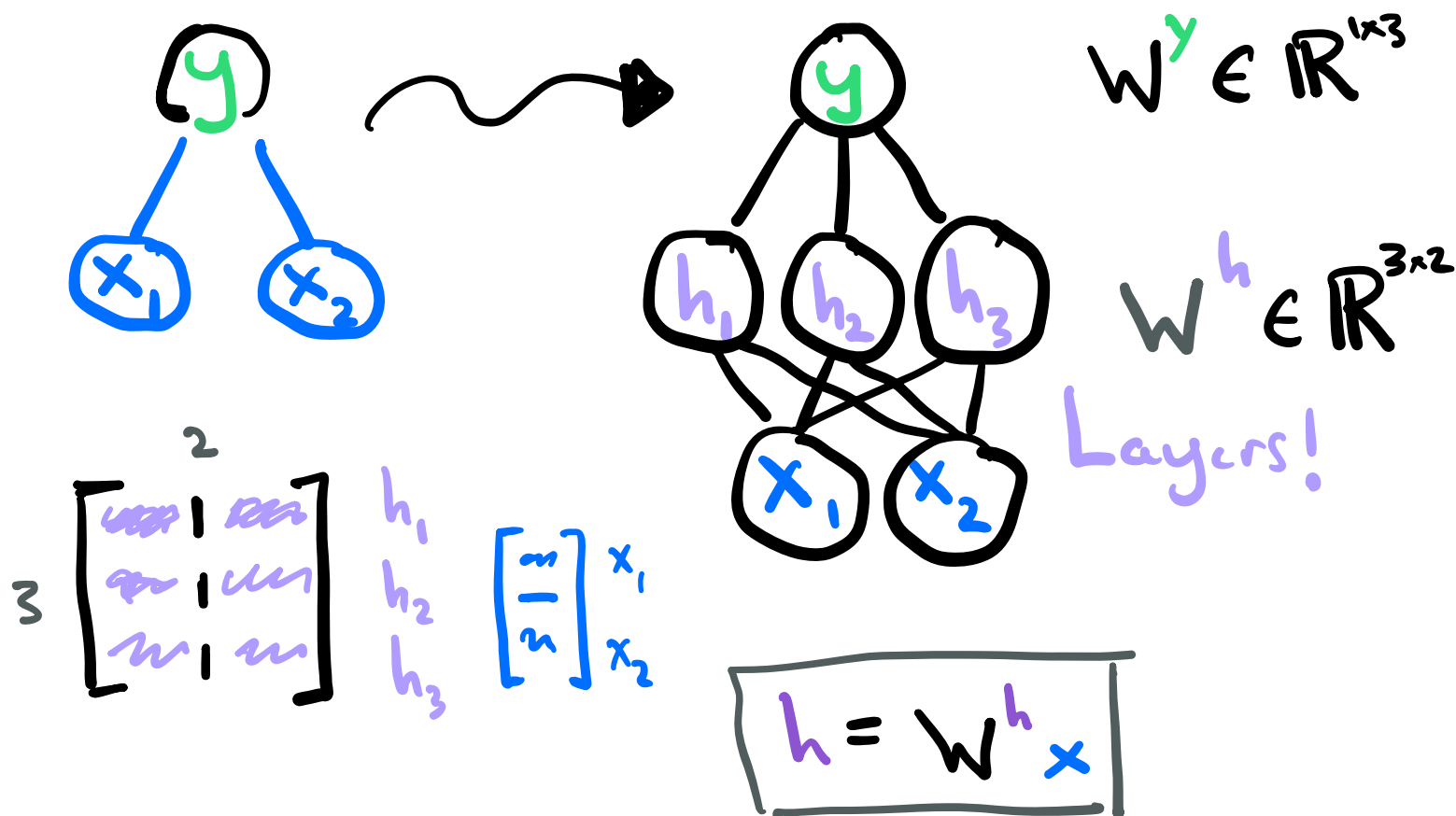
But consider:



$$\Delta = 1$$

$$\bigcirc = \emptyset$$

How can we address This?



BUT This is still linear!

$$y = W_1^y \cdot h_1 + W_2^y \cdot h_2 + W_3^y \cdot h_3$$

$$\hookrightarrow = W_{1:}^h \cdot x$$

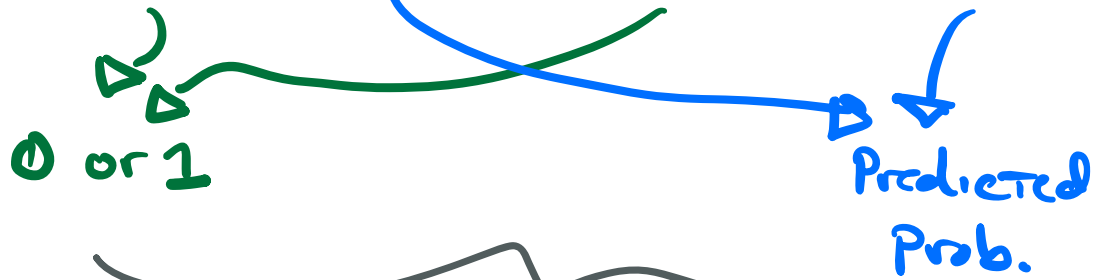
$$= W_1^y (W_{1:}^h \cdot x) + W_2^y (W_{2:}^h \cdot x) + W_3^y (W_{3:}^h \cdot x)$$

So just a weirdly parameterized

linear model! [See Colab NB.]

Quick aside: Binary Cross Entropy (BCE)

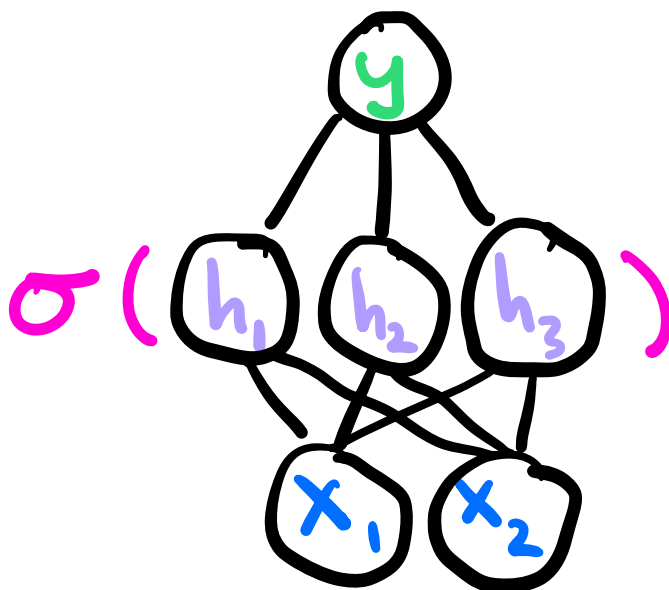
$$\text{Loss} = -[y \log \hat{y} + (1-y) \log(1-\hat{y})]$$



Want to be big so high $\hat{y}$
When $y=1$ and small otherwise

(As in your homework!)

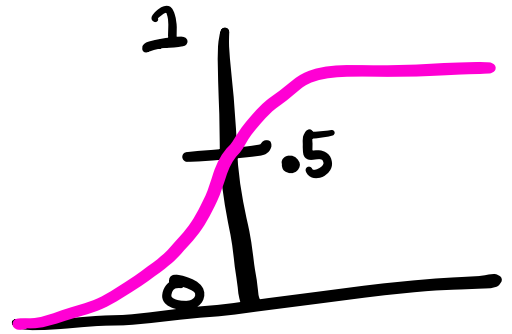
Enter activation functions σ



Applied
element
wise.

One choice for σ : Sigmoid

$$\sigma(x) = \frac{1}{1+e^{-x}}$$



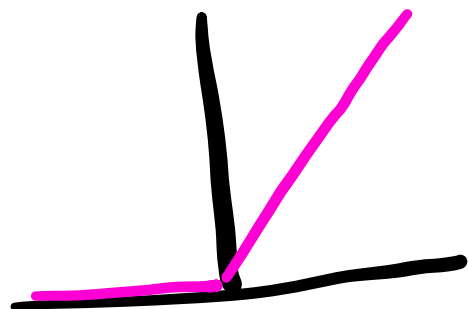
$$\frac{d}{dx} \sigma(x) = \sigma(x)(1 - \sigma(x))$$

Note SoftMax $\neq$ Sigmoid!

$$\boxed{\text{SM}(\underset{\substack{\uparrow \\ \text{vector}}}{z})_i = \frac{e^{z_i}}{\sum_j e^{z_j}}}$$

Rectified Linear Unit (ReLU)

$$\sigma(x) = \begin{cases} x & \text{if } x > 0 \\ 0 & \text{otherwise} \end{cases}$$

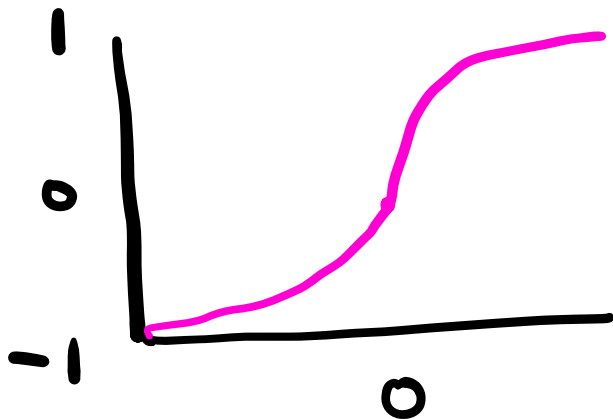


$$\frac{d}{dx} \sigma(x) = \begin{cases} 1 & \text{if } x > 0 \\ \emptyset & \text{otherwise} \end{cases}$$

Tanh

$$\sigma(x) = \frac{1 - \exp\{-2x\}}{1 + \exp\{-2x\}}$$

S-Shaped like Sigmoid but
ranges -1 to 1



$$\frac{d}{dx} \sigma(x) = 1 - \text{Tanh}^2(x)$$

(See notebook on activations)

The **hidden layer** as a learned representation

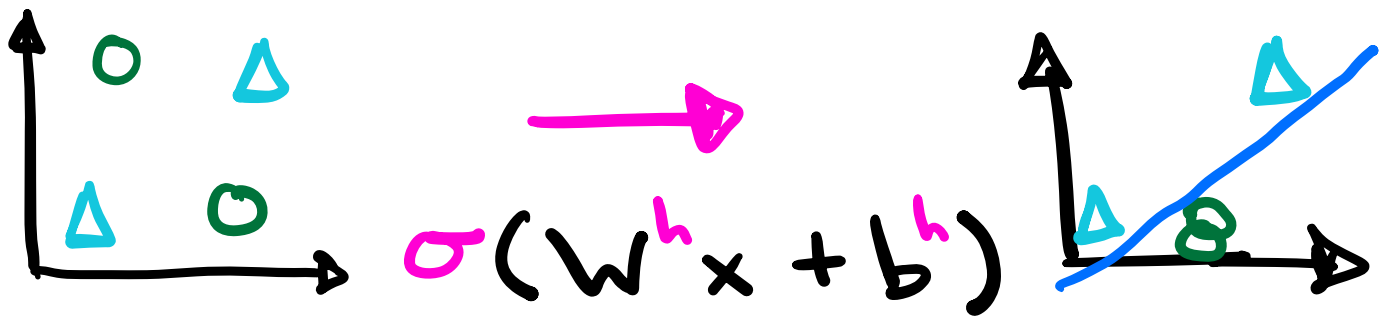
$$x_i \in \mathbb{R}^2$$

$$h_i = \sigma(W^h x) \in \mathbb{R}^2$$

$$\hat{y}_i = \sigma(\underbrace{W^y \cdot h_i}_{\text{Still a linear model!}})$$

Still a linear model!
Just over **h**

Consider XOR again.



Suppose σ is **ReLU** and:

$$W^h = \begin{bmatrix} 1 & 1 \\ 1 & 1 \end{bmatrix} \quad b^h = \begin{bmatrix} 0 \\ -1 \end{bmatrix}$$

$$x_1 \begin{bmatrix} 0 \\ 0 \end{bmatrix} \quad h_1 = \text{ReLU} \left(\begin{bmatrix} 0 \\ -1 \end{bmatrix} \right) = \begin{bmatrix} 0 \\ 0 \end{bmatrix}$$

$$x_2 \begin{bmatrix} 0 \\ 1 \end{bmatrix} \quad h_2 = \text{ReLU} \left(\begin{bmatrix} 1 \\ 0 \end{bmatrix} \right) = \begin{bmatrix} 1 \\ 0 \end{bmatrix}$$

$$x_3 \begin{bmatrix} 1 \\ 0 \end{bmatrix} \quad h_3 = \text{ReLU} \left(\begin{bmatrix} 2 \\ 0 \end{bmatrix} \right) = \begin{bmatrix} 1 \\ 0 \end{bmatrix}$$

$$x_4 \begin{bmatrix} 2 \\ 1 \end{bmatrix} \quad h_4 = \text{ReLU} \left(\begin{bmatrix} 2 \\ 1 \end{bmatrix} \right) = \begin{bmatrix} 2 \\ 1 \end{bmatrix}$$